

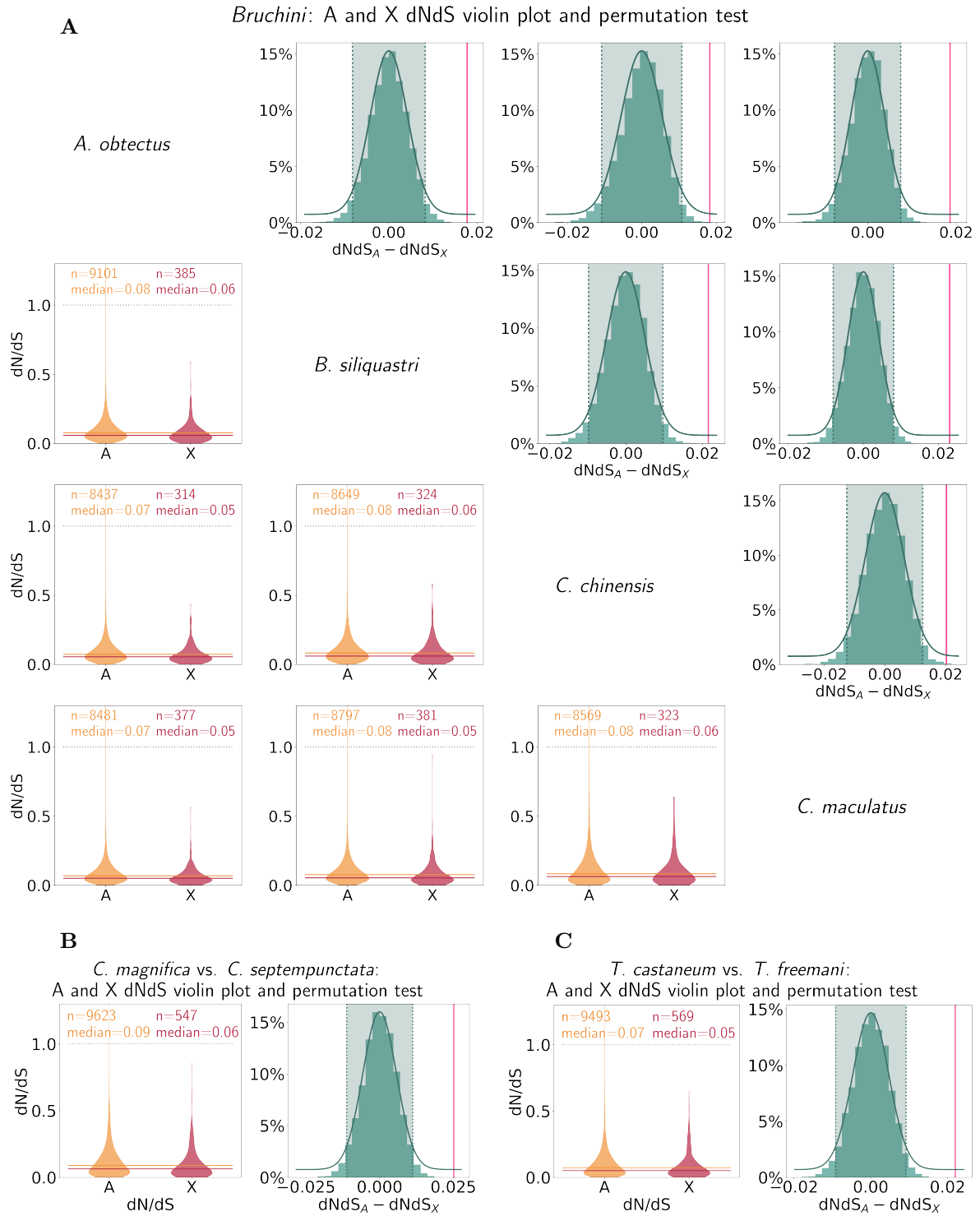


Figure 1: dN/dS of autosomal (A) and X-linked (X) genes Pairwise comparisons between members of three species groups with violin plots of dN/dS values for X-linked and autosomal 1-to-1 orthologs. Permutation tests (n=10000, visualized with histograms, pink line shows measured median difference) show that dN/dS is significantly lower for X-linked orthologs in all comparisons within *Bruchini* (A), *Coccinella* (B), and *Tribolium* (C).

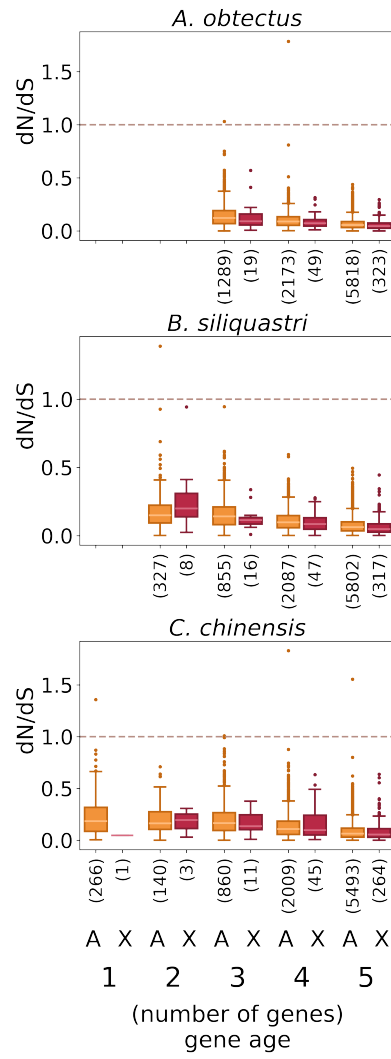


Figure 2: **Gene age:** Pairwise comparisons of *C. maculatus* with *A. obtectus*, *B. siliquastri* and *C. chinensis* respectively. The X-axis shows sample size in parentheses, as well as gene age (less conserved orthologs of age 1 to most conserved orthologs of age 5), and autosomal or X-linkage of orthologs (A or X). The box plots show dN/dS for autosomal (yellow) or X-linked (red) orthologs by gene age for *C. maculatus* compared to three other *Bruchini* species.

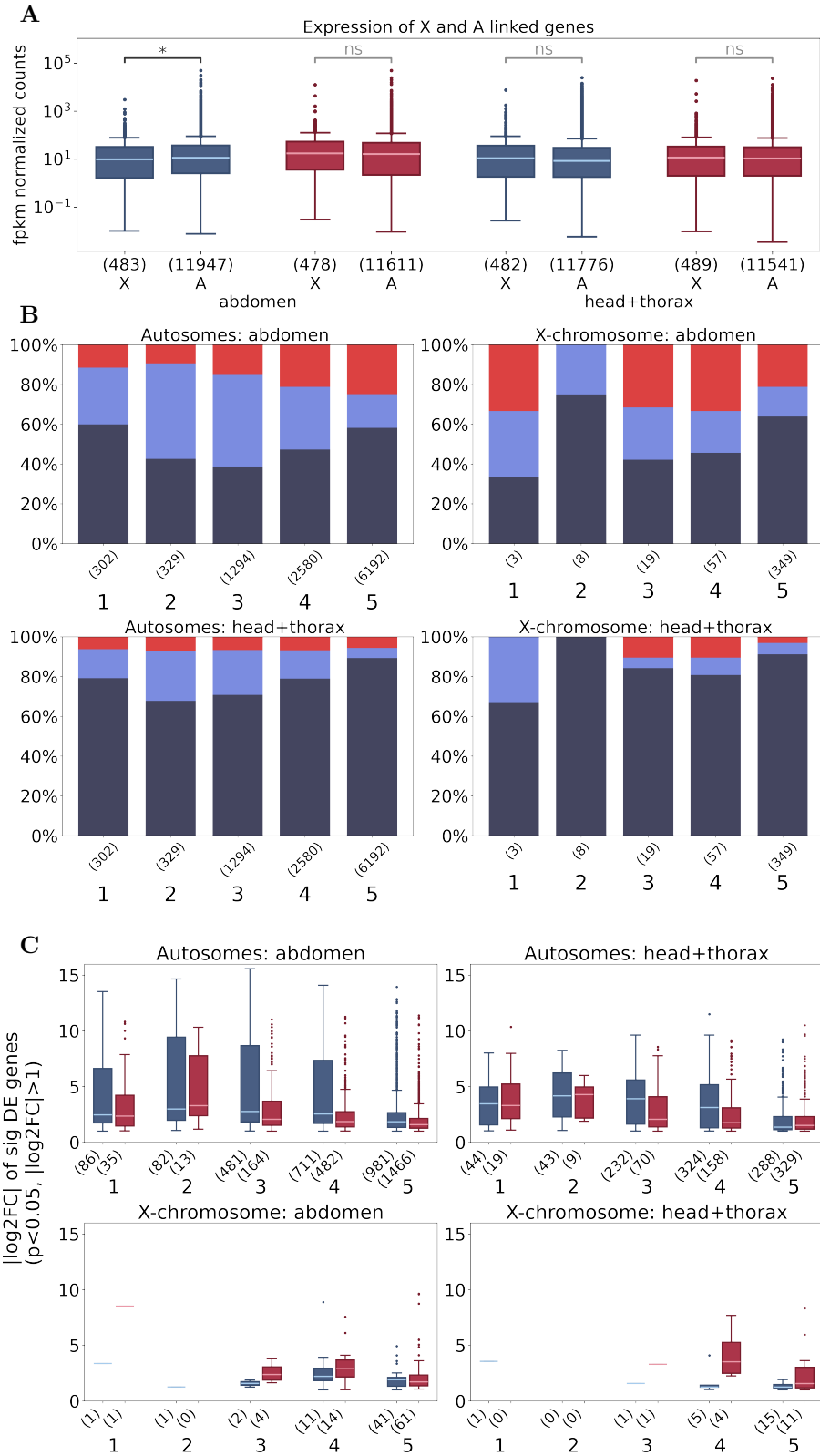


Figure 3: DC and gene age by sex-bias analysis: blue boxplots show male samples and red boxplots show female samples. X-axes ticks show sample size in parentheses, as well as gene age (less conserved orthologs of age 1 to most conserved orthologs of age 5), or autosomal or X-linkage of orthologs (A or X). **(A):** Gene expression (FPKM-normalized mean > 0 across three biological replicates) split by tissue and A or X-linkage. Mann-Whitney-U test for significant differences is applied and significant differences are marked with an asterisk. **(B):** proportion of sex-bias categories for each gene age category according to abdominal or head and thorax tissues, as well as autosomal or X-linked orthologs. Unbiased in grey, female-biased in red and male-biased in blue. **(C):** Absolute $\log_2FC$ for significantly male-biased (blue) or female-biased (red) orthologs of *C. maculatus* and *C. chinensis* for increasing gene age. Split by autosomal and X-linked orthologs, as well as $\log_2FC$ determined from abdominal tissue (reproductive) or head and thorax tissue (somatic).

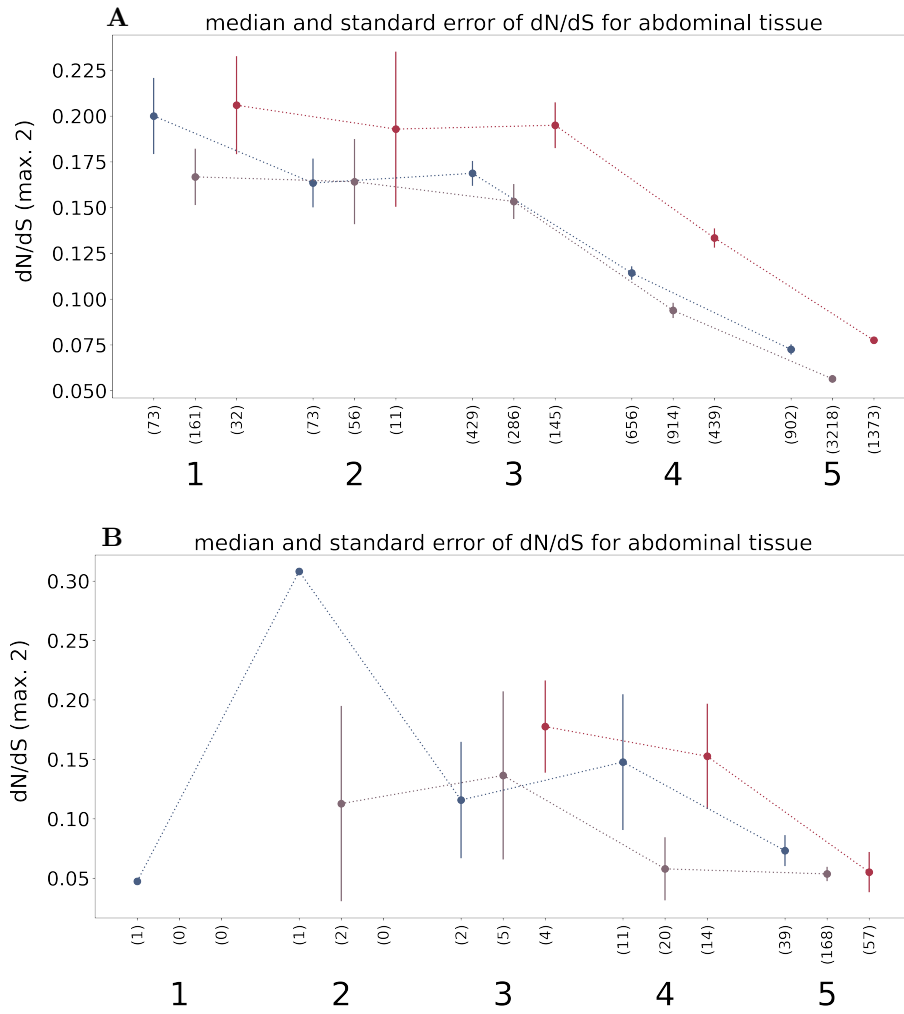


Figure 4: **Impact of gene age and sex-bias on dN/dS and positively selected genes:** X-axes ticks show sample size in parentheses, as well as gene age from less conserved orthologs (age 1) to highly conserved orthologs (age 5), or autosomal or X-linkage of orthologs (A or X). (**A,B**): median dN/dS and standard error of autosomal orthologs, split by gene age as well as sex-bias category according to abdominal tissues with male-bias being blue, unbiased being grey and female-bias being red. (**A**): autosomal, (**B**): X-linked.